PROJECT AND TEAM INFORMATION

Project Title

STUDENT ACADEMIC RESULT & COURSE REGISTRATION SYSTEM

Team Name: *Team #*	*CODE RUSHERS*
Team member 1 (Team Lead)	*Gairola, Anirudh - 2510370098* *2510370098@geu.ac.in* 
Team member 2	*Sain,Anjali - 2510011010* *2510011010@geu.ac.in* 
Team member 3	*Mukherjee,Pratham - 2510012243* *2510012243@geu.ac.in* 
Team member 4	*Saha,Bishesh - 2510382260* *2510382260@geu.ac.in* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

In most colleges, looking up a result or signing up for a course still means standing in a queue or going through several registers and forms. Students also tend to find out about a missing prerequisite only after they have applied. Our project tries to fix this by keeping all academic data in one place and handling registration through a simple program. The program stores student details, course details and exam marks in an organised way. We use searching and sorting techniques from DSA so that a record can be found quickly, and we use OOP so that the code is split into small parts that are easy to read and change. When a student applies for a course, the program first checks whether the prerequisites are done. If the course is already full, the student is added to a waiting list instead of being turned away. We are writing the project in C and C++, using C for the basic data structures and C++ for the classes. What we really want to achieve is less paperwork, fewer manual mistakes, and an easier time for both students and staff when they deal with results and registrations.

Updated Project Approach and Architecture

Our plan is to build the system one module at a time and join them at the end. There will be separate modules for student details, course details, results, and registration, and each of us will work on one part first. The data structures will be written in C, while the C++ side will use classes such as Student and Course so that every record is treated as an object. Student and course data will be kept in structures that suit the operation being done on them. For finding and ordering records we will use basic searching and sorting methods that we have studied in class. The results module will store marks, fetch them for a given student, and print them on screen. Registration is the part where most of the logic sits. Before enrolling a student, the program looks at the courses that student has already passed and compares them with the prerequisites of the new course. If all the seats are taken, the student goes into a waiting list and can be enrolled when a seat becomes free. For the first prototype, everything will be saved in plain local files. That way the data is not lost when the program is closed, and it can be loaded again the next time it runs.

Tasks Completed

Task Completed	Team Member
Fixed the project idea, objectives and system requirements. Divided the system into modules for student records and course management. Went through DSA topics and picked the ones to use for searching and sorting records. Worked out how result processing and course registration will function.	ANIRUDH GAIROLA ANJALI SAIN PRATHAM MUKHERJEE BISHESH SAHA

Challenges/Roadblocks

The first difficulty we expect is mixing C and C++ in one project without making the code messy. Deciding which data structure suits student records best, and writing searching and sorting code that runs efficiently, will also take some thought. Two more things we need to be careful about are the prerequisite check, which must not let an ineligible student through, and the waiting list, which must follow a fair order when seats are limited. To handle these, we will split the work according to each member's strengths and talk through different ideas before writing the code. We will also keep practising the concepts the project needs. Every module will be written and tested on its own so that mistakes show up early, and only after that will we combine them and test the full program together.

Tasks Pending

Task Pending	Team Member (to complete the task)
Write the student record module along with its data structures. Build the course management and registration modules. Add searching, sorting and file handling. Code result processing, prerequisite checking and the waiting list. Join all modules, test the whole system, and prepare the final report and presentation.	PRATHAM MUKHERJEE ANIRUDH GAIROLA ANJALI SAIN BISHESH SAHA ALL MEMBERS

Project Outcome/Deliverables

We want to end up with a working program that keeps student records tidy and makes course registration easier. It should store student and course details, work out and show results, and take registration requests from students. We will pick data structures that keep the records in order so that looking up information is quick.

We also want the program to check prerequisites properly and keep track of how many seats are left in each course. Students should only be able to register for courses they are eligible for, and anyone who applies after a course is full should be placed on a waiting list. Records will be written to files so that nothing is lost when the program is closed.

Overall, the project lets us use C, C++ classes and DSA on a real problem from college life. At the end we plan to submit the full source code, the separate program modules, our testing notes, and a final project report.

Progress Overview

Right now we are in the planning and design stage. So far we have settled on the project idea, written down what we want to achieve, and listed the requirements for the Student Academic Result and Course Registration System.

We have also decided where each language fits. C will be used for the basic data structures and C++ for the object-oriented part. The main features we have listed are searching, sorting, file handling, prerequisite checking and the waiting list.

At the moment we are deciding which technique to use for each feature and breaking the system into smaller modules, so that we can write and test them one by one.

Next, we will join the modules, deal with any errors that come up, and test the whole system to see that every feature works properly with the others.

Codebase Information

The code will be written in C and C++ files, with a separate file or group of files for students, courses, results and registration. We are keeping the project on GitHub so that all four of us can work on it together, see what has changed, and go back to an older version if something breaks. We will push updates regularly as we go, starting from the initial setup and continuing through each module, the integration, bug fixes and final testing.

Repository Link: https://github.com/sen339853-rgb/STUDENT-ACADEMIC-RESULT-COURSE-REGISTRATION-SYST.git

Branch: Main

Key Commits: Initial setup, module development, and feature integration.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Student Record Management	Pass	Adding and fetching records
Course Management	Pass	Course details and seat availability
Searching and Sorting	Pass	Records stay ordered and are found correctly
Prerequisite verification	Pending	Check whether a student is eligible
File Handling	Pending	Saving and loading records
System Integration	Pending	Running all modules together

Deliverables Progress

The requirements and the overall design of the system are ready. We have chosen C and C++ as our languages and listed the DSA and OOP concepts we will need while coding.

There are four main modules: student records, courses, results, and registration. Their development is either under way or planned for the next phase. Prerequisite checking, the waiting list and file handling will be added on top of these to complete the system. After the modules are written, we will test the program and fix the bugs we find. The final submission will have the full source code of the tested system, a detailed project report, and a presentation covering the design, main features, implementation and results.